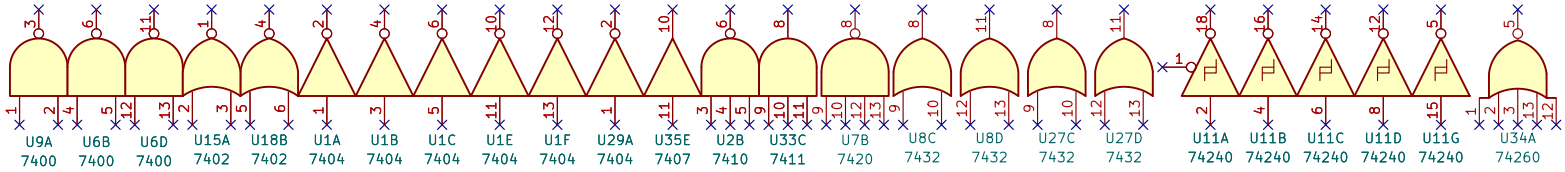
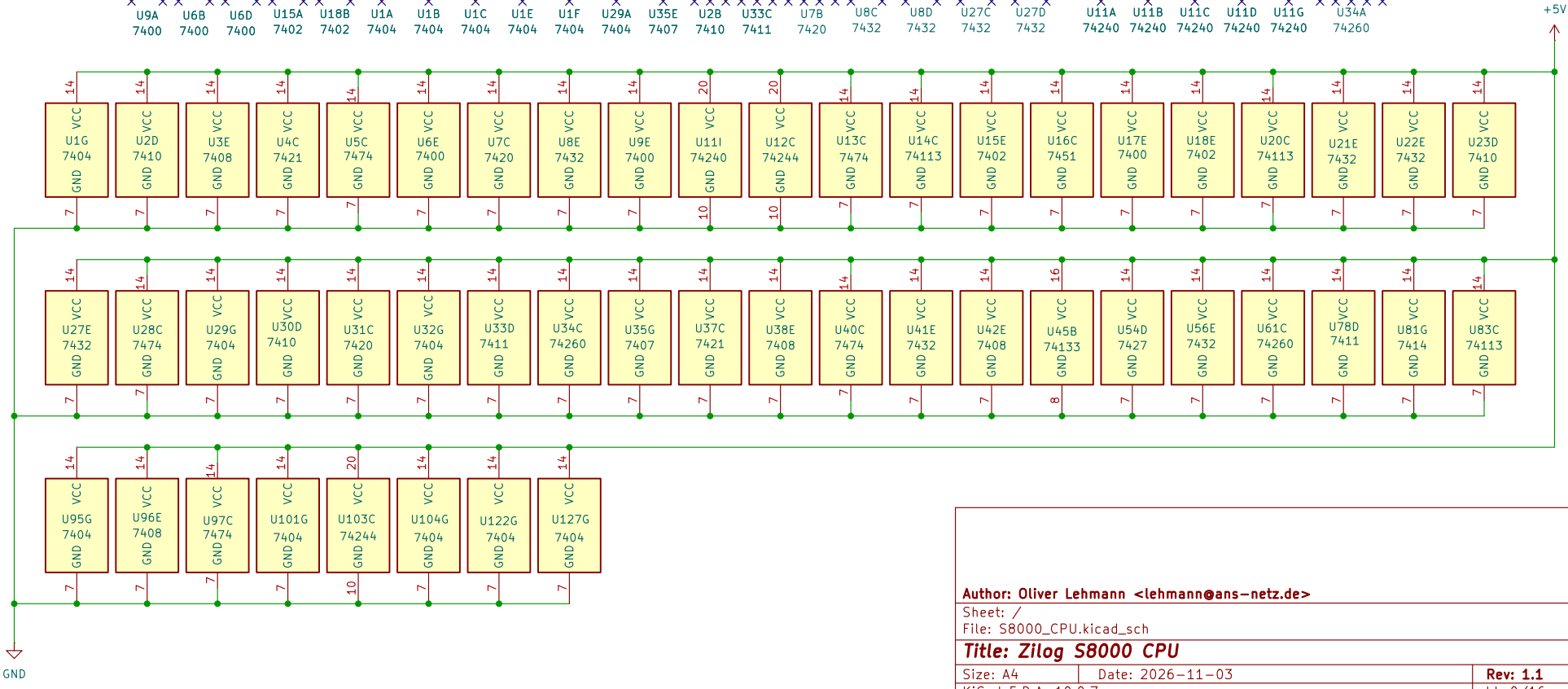
S8000 CPU 01 S8000 CPU 02 S8000 CPU 03 S8000 CPU 04 S8000 CPU 05 S8000 CPU 06 S8000 CPU 07 S8000 CPU 08 S8000 CPU 09 S8000 CPU 10 S8000 CPU 11 S8000 CPU 12 S8000 CPU 13 S8000 CPU 14 S8000 CPU 15





Author: Oliver Lehmann <lehmann@ans-netz.de>

Sheet: /
File: S8000_CPU.kicad_sch

Title: Zilog S8000 CPU

Size: A4	Date: 2026-11-03	**Rev: 1.1**
KiCad E.D.A. 10.0.7		Id: 0/16

I.C.	+5V (PIN)	GND	−5V (PIN)	DESIGNATIONS	SPARE GATES
74S00	14	7		U6,9,17	U9 − A,D, U6 − B,D
74S02	14	7		U15	U15 − A
74LS02	14	7		U18	U18 − B
74S04	14	7		U1,29,95	U1 − A−C,E,F, U29 − A
74LS04	14	7		U32,101,104,122, 127	
7407	14	7		U35	U35 − E
74S08	14	7		U3,96	
74LS08	14	7		U38,42	
74S10	14	7		U2,23,30	U2 − B
74S11	14	7		U78	
74LS11	14	7		U33	U33 − C
74LS14	14	7		U81	
74S20	14	7		U7,31	U7 − B
74LS21	14	7		U4,37	
74LS27	14	7		U54	
74S32	14	7		U21	
74LS32	14	7		U8,22,27,41	U8 − C,D, U27 − C,D
74S51	14	7		U16	
74S74	14	7		U5,13,97	
74LS74	14	7		U28,40	
74S85	14	7		U52,53	
74S113 1)	14	7		U14,20,83	
74LS133	16	8		U45	
74LS138 2)	16	8		U24−26,39,55	
74LS175	16	8		U36	

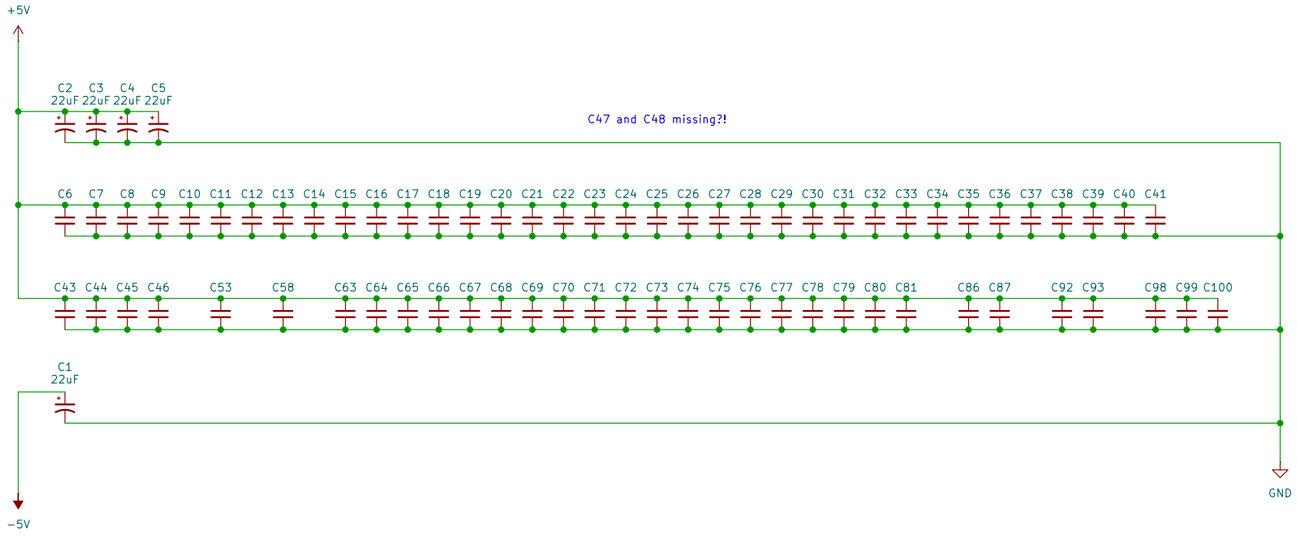
1) U83 is a LS on my board

2) U25 and U26 are both S on my board

I.C. TYPE	+5V (PIN)	GND	−5V (PIN)	DESIGNATIONS	SPARE GATES
74S240	20	10		U11,68,109	
74S241	20	10		U99	
74LS244	20	10		U12,51,67,73,103, 108,110,112,114− 118	
74LS245	20	10		U46,63.111,113	
74S260	14	7		U34	U34 − A
74LS260	14	7		U61	~~U61 − B~~
74S373	20	10		U47,62,71,72	
74LS374	20	10		U48,49,50,65,66	
26LS30	1	5	8	U100,102,105,124, 126,128	
26LS32	16	8		U106,107,120,121, 123,125	
93S16	16	8		U98	
Z80B−CTC	25	5		U43,44,89	
2114(p)−3	18	9		U57−60	
Z8001A−CPU	11	36		U64	
2732	24	12		U74−77	
Z80CLK DRV	1	5		U79,80	
2.4576 MHz	14	7		U84	
44.440 MHz	14	7		U85	
Z8010A MMU	11	35		U86,87,88	
Z80B−SIO/2	9	31		U90−93	
Z80B−PIO	26	11		U94	

REF DESIGNATION	
LAST USED	NOT USED
C100	–––
CR1	–––
DS4	–––
R20	–––
U128	–––

NOTES: UNLESS OTHERWISE SPECIFIED
1. ALL RESISTORS ARE 1Ka, 1/4W, 5%
2. CAPACITOR VALUES ARE IN MICROFARADS.
3. ALPHANUMERIC DENOTES GATES

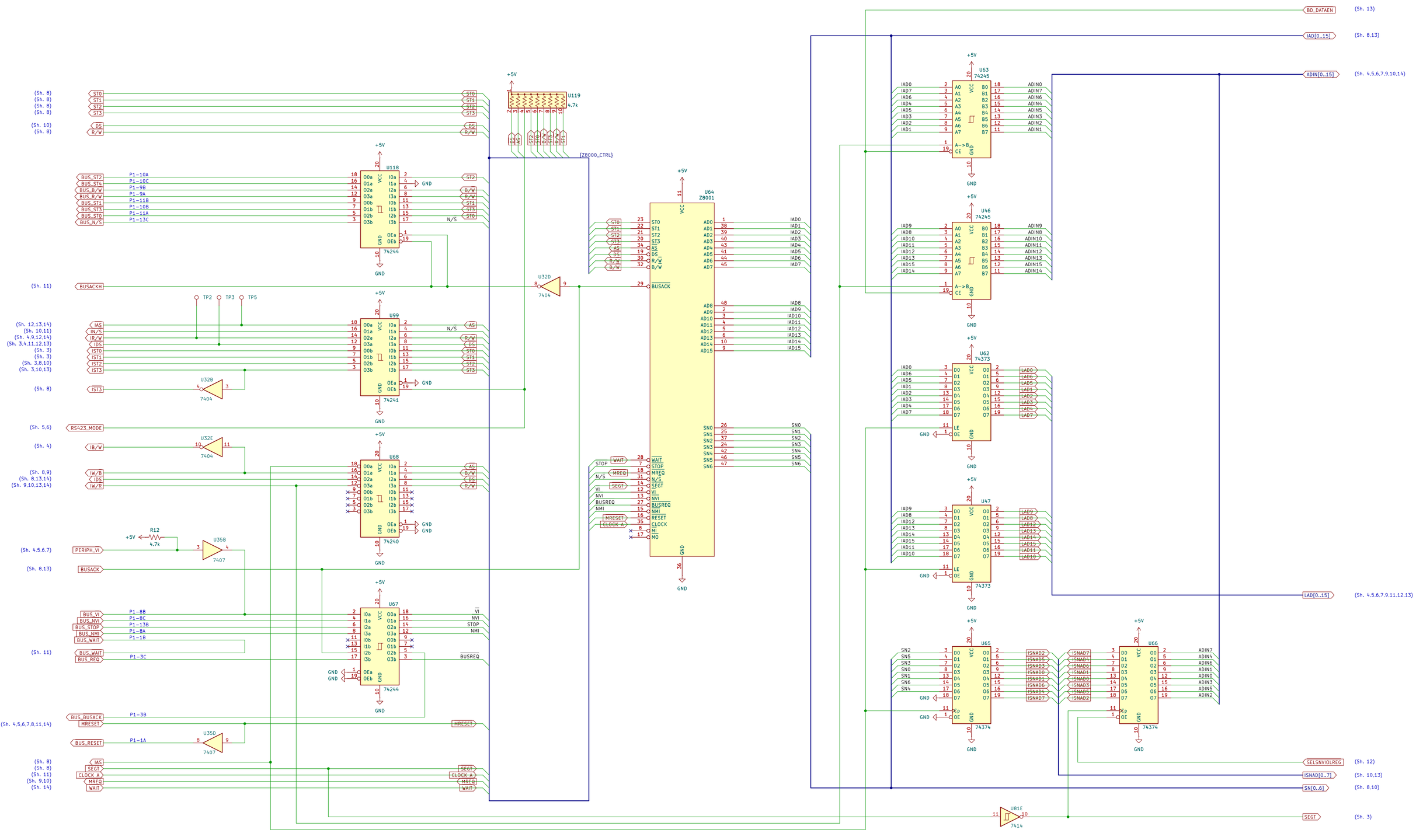


Author: Oliver Lehmann <lehmann@ans−netz.de>

Sheet: /S8000 CPU 01/
File: S8000_CPU_01.kicad_sch

Title: Zilog S8000 CPU

Size: A3	Date: 2026−11−03	**Rev: 1.1**
KiCad E.D.A. 10.0.7		Id: 1/16



Z8001A CPU
Segment Violation Register U66
Author: Oliver Lehmann <lehmann@ans-netz.de>
Sheet: /S8000 CPU 02/
File: S8000_CPU_02.kicad_sch

Title: Zilog S8000 CPU

Size: A2	Date: 2026-11-03	**Rev: 1.1**
KiCad E.D.A. 10.0.7		Id: 2/16

(Sh. 2) IAD[0..15]

(Sh. 12) IF1LOBYTE

(Sh. 2) SEGT

(Sh. 12) TRPLOBYTE

ADIN[0..15] (Sh. 13)

U48 74374 — +5V, GND

U49 74374 — +5V, GND

U50 74374 — +5V, GND

IAD7 IAD6 IAD5 IAD4 IAD1 IAD2 IAD3 IAD0

ADIN7 ADIN6 ADIN5 ADIN4 ADIN3 ADIN2 ADIN1 ADIN0

U27A 7432

(Sh. 2) IST0
(Sh. 2) IST1
(Sh. 2) IST2
(Sh. 2) IST3

U25 74138 — +5V, GND

R4 1k

U26 74138 — +5V, GND

R2 1k

R3 1k

U4A 7421

I/O_REF (Sh. 11.12)

SEGSTACK (Sh. 8)

VIACK (Sh. 12)

MMU/EPUREF (Sh. 13)

STACKMEMREQ (Sh. 4)

INSTFETCH1STWD (Sh. 8)

EPUXFER (Sh. 11)

U9C 7400

P2_NMI_SWITCH_NO — P2-26C

U9D 7400

P2_NMI_SWITCH_NC — P2-25C

(Sh. 4) MANUAL_NMI

(Sh. 4) LATCHECCERROR
(Sh. 10.11) POWERFAIL

U54B 7427

U35C 7407

TP6

P1-8A BUS_NMI

U8A 7432

ONBDNMI (Sh. 12)

U8B 7432

(Sh. 2) IDS

ENAONBDNMI (Sh. 10)

External Violation Registers (U48,49,50)
VIACK- generation
Author: Oliver Lehmann <lehmann@ans-netz.de>
Sheet: /S8000 CPU 03/
File: S8000_CPU_03.kicad_sch

Title: Zilog S8000 CPU

Size: A2	Date: 2026-11-03	**Rev: 1.1**
KiCad E.D.A. 10.0.7		Id: 3/16

(Sh. 2) ADIN[0..15]

+5V

U44
Z80 CTC

ADIN0 25 D0
ADIN1 26 D1
ADIN2 27 D2
ADIN3 28 D3
ADIN4 1 D4
ADIN5 2 D5
ADIN6 3 D6
ADIN7 4 D7

(Sh. 12) CTC 0 CE → 16 CE
18 CS0
19 CS1
14 M1
10 IOREQ
6 RD
12 INT
17 RESET
CTC 0

U41A
7432
(Sh. 2) IDS → 1
(Sh. 3) STACKMEMREQ → 2
3 → 15 CLK

23 CLK/TRG0
22 CLK/TRG1
21 CLK/TRG2
20 CLK/TRG3

+5V → 13 IEI
ZC/TO0 7
ZC/TO1 8
ZC/TO2 9
IEO 11

TP8

BAUD 0 (Sh. 5)
BAUD 1 (Sh. 5)
BAUD 2 (Sh. 5)
CTC 0 IEO (Sh. 14)

24 VCC
5 GND
GND

+5V

U43
Z80 CTC

ADIN0 25 D0
ADIN1 26 D1
ADIN2 27 D2
ADIN3 28 D3
ADIN4 1 D4
ADIN5 2 D5
ADIN6 3 D6
ADIN7 4 D7

TP10

(Sh. 12) CTC 1 CE → 16 CE
(Sh. 2) LAD1 → 18 CS0
(Sh. 2) LAD2 → 19 CS1
(Sh. 12) M1 → 14 M1
(Sh. 12) IOREQ → 10 IOREQ
(Sh. 12) RD → 6 RD
(Sh. 2) PERIPH_WI → 12 INT
(Sh. 2) MRESET → 17 RESET
CTC 1

(Sh. 11) CLOCK_B → 15 CLK
(Sh. 11) BAUD_CLK → 23 CLK/TRG0
22 CLK/TRG1
21 CLK/TRG2
20 CLK/TRG3

13 IEI
ZC/TO0 7
ZC/TO1 8
ZC/TO2 9
IEO 11

BAUD 3 (Sh. 5)
BAUD 4 (Sh. 5)
BAUD 5 (Sh. 5)
CTC 1 IEO (Sh. 7.14)

24 VCC
5 GND
GND

U33A
7411
(Sh. 2) IR/W → 1
(Sh. 9) LADD → 2
(Sh. 2) IB/W → 13
12

ENA BYTE SWAP (Sh. 13)

U32A
7404
1 → 2
ENA BYTE SWAP (Sh. 13)

(Sh. 14) SYS_DS → 4

GND → 3 J
1 PRE
5 Q
(Sh. 14) CLRPARITY → 1 K
2 CLR
6 Q̄

U14A
74113

(Sh. 14) BUS_ME → P1-12A → U81A 1 → 2
7414

LATCHECCERROR (Sh. 3.10,11)

Serial Input/Output: Baud Rate Generators
Byte Swap Buffer: create the input buffer enable signal
Parity Error Checking: parity error flip-flop
Author: Oliver Lehmann <lehmann@ans-netz.de>

Sheet: /S8000 CPU 04/
File: S8000_CPU_04.kicad_sch

Title: Zilog S8000 CPU

Size: A2	Date: 2026-11-03	**Rev: 1.1**
KiCad E.D.A. 10.0.7		Id: 4/16

(Sh. 2) ADIN[0..15]

(Sh. 12) SIO 0 CE

(Sh. 7.14) CTC 2 IEO

(Sh. 4) BAUD 0

(Sh. 4) BAUD 1

+5V

U93
Z80 SIO/2

ADIN0 40 D0
ADIN1 1 D1
ADIN2 39 D2
ADIN3 2 D3
ADIN4 38 D4
ADIN5 3 D5
ADIN6 37 D6
ADIN7 4 D7

9 VCC

34 B/A
33 C/D

35 CE
21 RESET
8 M1
36 IORQ
32 RD

20 CLK

5 INT
6 IEI
7 IEO

13 RxCA
14 TxCA

28 RxCB
27 TxCB

SIO 0

10 W/RDYA
16 DTRA
17 RTSA
15 TxDA
19 DCDA
18 CTSA
12 RxDA
11 SYNCA
30 W/RDYB
25 DTRB
24 RTSB
26 TxDB
22 DCDB
23 CTSB
29 RxDB

31 GND

GND

U127A 1 2 7404
U127E 11 10 7404
U127D 9 8 7404
U127C 5 6 7404
U127B 3 4 7404
U127F 13 12 7404

U128A 2 15 26LS30
U128C 6 11 26LS30
U128D 7 10 26LS30
U128B 3 14 26LS30

C94 C
C96 C
C97 C
C91 C
C88 C
C95 C

U107D 13 15 26LS32
U107A 3 1 26LS32
U107C 11 9 26LS32
U106C 11 9 26LS32
U106D 13 15 26LS32
U107B 5 7 26LS32

P2 – 4A P2_SIO_DSR0
P2 – 2B P2_SIO_CTS0
P2 – 1B P2_SIO_RXD0
P2 – 3B P2_SIO_DTR0
P2 – 3A P2_SIO_RTS0
P2 – 2A P2_SIO_TXD0
P2 – 1A P2_SIO_TXRTN0

P2 – 7A P2_SIO_DSR1
P2 – 5B P2_SIO_CTS1
P2 – 4B P2_SIO_RXD1
P2 – 6B P2_SIO_DTR1
P2 – 6A P2_SIO_RTS1
P2 – 5A P2_SIO_TXD1
P2 – 14C P2_SIO_TXRTN1

SIO 0 IEO (Sh. 14)

(Sh. 2) LAD1
(Sh. 2) LAD2

(Sh. 12) SIO 1 CE
(Sh. 12) MRESET
(Sh. 12) M1
(Sh. 12) IOREQ
(Sh. 12) RD

(Sh. 11) CLOCK B

(Sh. 2) PERIPH_VI

(Sh. 4) BAUD 2

(Sh. 4) BAUD 3

+5V

U92
Z80 SIO/2

ADIN0 40 D0
ADIN1 1 D1
ADIN2 39 D2
ADIN3 2 D3
ADIN4 38 D4
ADIN5 3 D5
ADIN6 37 D6
ADIN7 4 D7

9 VCC

34 B/A
33 C/D

35 CE
21 RESET
8 M1
36 IORQ
32 RD

20 CLK

5 INT
6 IEI
7 IEO

13 RxCA
14 TxCA

28 RxCB
27 TxCB

SIO 1

10 W/RDYA
16 DTRA
17 RTSA
15 TxDA
19 DCDA
18 CTSA
12 RxDA
11 SYNCA
30 W/RDYB
25 DTRB
24 RTSB
26 TxDB
22 DCDB
23 CTSB
29 RxDB

31 GND

GND

U104B 3 4 7404
U104A 1 2 7404
U104E 11 10 7404
U104D 9 8 7404
U104C 5 6 7404
U104F 13 12 7404

U126B 3 14 26LS30
U105A 2 15 26LS30
U126C 6 11 26LS30
U125D 13 15 26LS32
U106B 5 7 26LS32
U106A 3 1 26LS32

C89 C
C59 C
C90 C
C62 C
C60 C
C61 C

U105D 7 10 26LS30
U105B 3 14 26LS30
U105C 6 11 26LS30
U125A 3 1 26LS32
U125B 5 7 26LS32
U125C 11 9 26LS32

P2 – 10A P2_SIO_DSR2
P2 – 8B P2_SIO_CTS2
P2 – 7B P2_SIO_RXD2
P2 – 9B P2_SIO_DTR2
P2 – 9A P2_SIO_RTS2
P2 – 8A P2_SIO_TXD2
P2 – 18C P2_SIO_TXRTN2

P2 – 13A P2_SIO_DSR3
P2 – 11B P2_SIO_CTS3
P2 – 10B P2_SIO_RXD3
P2 – 12B P2_SIO_DTR3
P2 – 12A P2_SIO_RTS3
P2 – 11A P2_SIO_TXD3
P2 – 26A P2_SIO_TXRTN3

SIO 1 IEO (Sh. 14)

−5V +5V

U105E 26LS30
U126E 26LS30
U128E 26LS30
U106E 26LS32
U107E 26LS32
U125E 26LS32

(Sh. 2) RS423_MODE

GND

Serial Input/Output: SIO channels 0 through 3
Author: Oliver Lehmann <lehmann@ans-netz.de>
Sheet: /S8000 CPU 05/
File: S8000_CPU_05.kicad_sch

Title: Zilog S8000 CPU

Size: A2	Date: 2026-11-03	**Rev: 1.1**
KiCad E.D.A. 10.0.7		Id: 5/16

(Sh. 2) ADIN[0..15]

(Sh. 12) SIO 2 CE

(Sh. 14) SIO 2 IEI

(Sh. 4) BAUD 4

(Sh. 4) BAUD 5

+5V

U91
Z80 SIO/2

ADIN0 40 D0
ADIN1 1 D1
ADIN2 39 D2
ADIN3 2 D3
ADIN4 38 D4
ADIN5 3 D5
ADIN6 37 D6
ADIN7 4 D7

9 VCC

34 B/A
33 C/D

35 CE
21 RESET
8 M1
36 IORQ
32 RD

20 CLK

5 INT
6 IEI
7 IEO

13 RxCA
14 TxCA

28 RxCB
27 TxCB

10 W/RDYA
16 DTRA
17 RTSA
15 TxDA
19 DCDA
18 CTSA
12 RxDA
11 SYNCA
30 W/RDYB
25 DTRB
24 RTSB
26 TxDB
22 DCDB
23 CTSB
29 RxDB

SIO 2

31 GND

GND

U122F 13 12 7404
U122C 5 6 7404
U122D 9 8 7404

U124A 2 15 26LS30
U124C 6 11 26LS30
U124D 7 10 26LS30

C82 C
C84 C
C85 C

U123A 3 2 1 26LS32
U123C 11 10 9 26LS32
U123D 13 14 15 26LS32

P2 – 16A P2_SIO_DSR4
P2 – 14B P2_SIO_CTS4
P2 – 13B P2_SIO_RXD4
P2 – 15B P2_SIO_DTR4
P2 – 15A P2_SIO_RTS4
P2 – 14A P2_SIO_TXD4
P2 – 27A P2_SIO_TXRTN4

U122B 3 4 7404
U122A 1 2 7404
U122E 11 10 7404

U102B 3 13 14 26LS30
U102D 7 9 10 26LS30
U124B 3 13 14 26LS30

C55 C
C57 C
C83 C

U121D 13 14 15 26LS32
U121C 11 10 9 26LS32
U123B 5 6 7 26LS32

P2 – 19A P2_SIO_DSR5
P2 – 17B P2_SIO_CTS5
P2 – 16B P2_SIO_RXD5
P2 – 18B P2_SIO_DTR5
P2 – 18A P2_SIO_RTS5
P2 – 17A P2_SIO_TXD5
P2 – 25B P2_SIO_TXRTN5

SIO 2 IEO (Sh. 14)

(Sh. 2) LAD1
(Sh. 2) LAD2

(Sh. 12) SIO 3 CE
(Sh. 12) MRESET
(Sh. 12) M1
(Sh. 12) IOREQ
(Sh. 12) RD

(Sh. 11) CLOCK B

(Sh. 2) PERIPH_VI
(Sh. 14) SIO 3 IEI

(Sh. 7) BAUD 6

(Sh. 7) BAUD 7

+5V

U90
Z80 SIO/2

ADIN0 40 D0
ADIN1 1 D1
ADIN2 39 D2
ADIN3 2 D3
ADIN4 38 D4
ADIN5 3 D5
ADIN6 37 D6
ADIN7 4 D7

9 VCC

34 B/A
33 C/D

35 CE
21 RESET
8 M1
36 IORQ
32 RD

20 CLK

5 INT
6 IEI
7 IEO

13 RxCA
14 TxCA

28 RxCB
27 TxCB

10 W/RDYA
16 DTRA
17 RTSA
15 TxDA
19 DCDA
18 CTSA
12 RxDA
11 SYNCA
30 W/RDYB
25 DTRB
24 RTSB
26 TxDB
22 DCDB
23 CTSB
29 RxDB

SIO 3

31 GND

GND

U101C 5 6 7404
U101F 13 12 7404
U101E 11 10 7404

U100C 6 11 26LS30
U102A 2 15 26LS30
U102C 6 11 26LS30

C51 C
C54 C
C56 C

U120D 13 14 15 26LS32
U121B 5 6 7 26LS32
U121A 3 2 1 26LS32

P2 – 22A P2_SIO_DSR6
P2 – 20B P2_SIO_CTS6
P2 – 19B P2_SIO_RXD6
P2 – 21B P2_SIO_DTR6
P2 – 21A P2_SIO_RTS6
P2 – 20A P2_SIO_TXD6
P2 – 26B P2_SIO_TXRTN6

U101A 1 2 7404
U101B 3 4 7404
U101D 9 8 7404

U100A 2 15 26LS30
U100B 3 13 14 26LS30
U100D 7 9 10 26LS30

C49 C
C50 C
C52 C

U120A 3 2 1 26LS32
U120B 5 6 7 26LS32
U120C 11 10 9 26LS32

P2 – 25A P2_SIO_DSR7
P2 – 23B P2_SIO_CTS7
P2 – 22B P2_SIO_RXD7
P2 – 24B P2_SIO_DTR7
P2 – 24A P2_SIO_RTS7
P2 – 23A P2_SIO_TXD7
P2 – 27B P2_SIO_TXRTN7

SIO 3 IEO (Sh. 14)

−5V +5V

U100E 26LS30 4 MODE VEE 8 5 GND
U102E 26LS30 4 MODE VEE 8 5 GND
U124E 26LS30 4 MODE VEE 8 5 GND
U120E 26LS32 4 VCC 16 G 8 GND
U121E 26LS32 4 VCC 16 G 8 GND
U123E 26LS32 4 VCC 16 G 8 GND

(Sh. 2) RS423_MODE

GND

Serial Input/Output: SIO channels 4 through 7
Author: Oliver Lehmann <lehmann@ans-netz.de>
Sheet: /S8000 CPU 06/
File: S8000_CPU_06.kicad_sch

Title: Zilog S8000 CPU

Size: A2	Date: 2026-11-03	**Rev: 1.1**
KiCad E.D.A. 10.0.7		Id: 6/16

(Sh. 2) ADIN[0..15]

(Sh. 12) CTC 2 CE

(Sh. 2) MRESET

(Sh. 11) BAUD CLK

(Sh. 4) CTC 1 IEO

U89
Z80 CTC
CTC 0

+5V

ADIN0 ADIN1 ADIN2 ADIN3 ADIN4 ADIN5 ADIN6 ADIN7

CE CS0 CS1 M1 IOREQ RD INT RESET

CLK CLK/TRG0 CLK/TRG1 CLK/TRG2 CLK/TRG3 IEI

ZC/TO0 ZC/TO1 ZC/TO2 IEO

VCC GND

BAUD 6 (Sh. 6)
BAUD 7 (Sh. 6)
CTC 2 IEO (Sh. 5)

A[0..7]

B[0..7]

U94
Z80 PIO

+5V

ADIN0 ADIN1 ADIN2 ADIN3 ADIN4 ADIN5 ADIN6 ADIN7

B/A C/D CE M1 IORQ RD CLK INT IEI IEO

PA0 PA1 PA2 PA3 PA4 PA5 PA6 PA7

PB0 PB1 PB2 PB3 PB4 PB5 PB6 PB7

ARDY ASTB BRDY BSTB

VCC GND

A1 A2 A3 A5 A6 A7

B0 B1 B2 B3 B4 B5 B6 B7

(Sh. 2) LAD1
(Sh. 2) LAD2

(Sh. 12) PIO CE

(Sh. 12) M1
(Sh. 12) IOREQ
(Sh. 12) RD

(Sh. 11) CLOCK B

(Sh. 2) PERIPH_VI
(Sh. 14) PIO_0_IEI

U42B
7408

Data Products Interface
E16 E17
E17_DEF
E18
Centronics Interface

U11E
74240

U103A
74244

U103B
74244

A3 A1 A2 A6 A5 A7

I0a I1a I2a I3a I0b I1b I2b I3b

O0a O1a O2a O3a O0b O1b O2b O3b

OEa OEb GND

P2 – 12C P2_PIO_DSTRB/DSTRB
P2 – 13C P2_PIO_N.U./INPUT PRIME

R20 1k +5V
R19 1k +5V

P2 – 20C P2_PIO_IFVALID/FAULT
P2 – 21C P2_PIO_ON-LINE/SELECT

R5 1k +5V
P2 – 17C P2_PIO_BUSY

U108
74244

B0 B1 B2 B3 B4 B5 B6 B7

P2 – 1C P2_PIO_DATA0
P2 – 2C P2_PIO_DATA1
P2 – 3C P2_PIO_DATA2
P2 – 5C P2_PIO_DATA3
P2 – 6C P2_PIO_DATA4
P2 – 8C P2_PIO_DATA5
P2 – 9C P2_PIO_DATA6
P2 – 10C P2_PIO_DATA7

Data Products Interface
E14 E15
E14_DEF
E13
Centronics Interface

U11F
74240

U128
74244

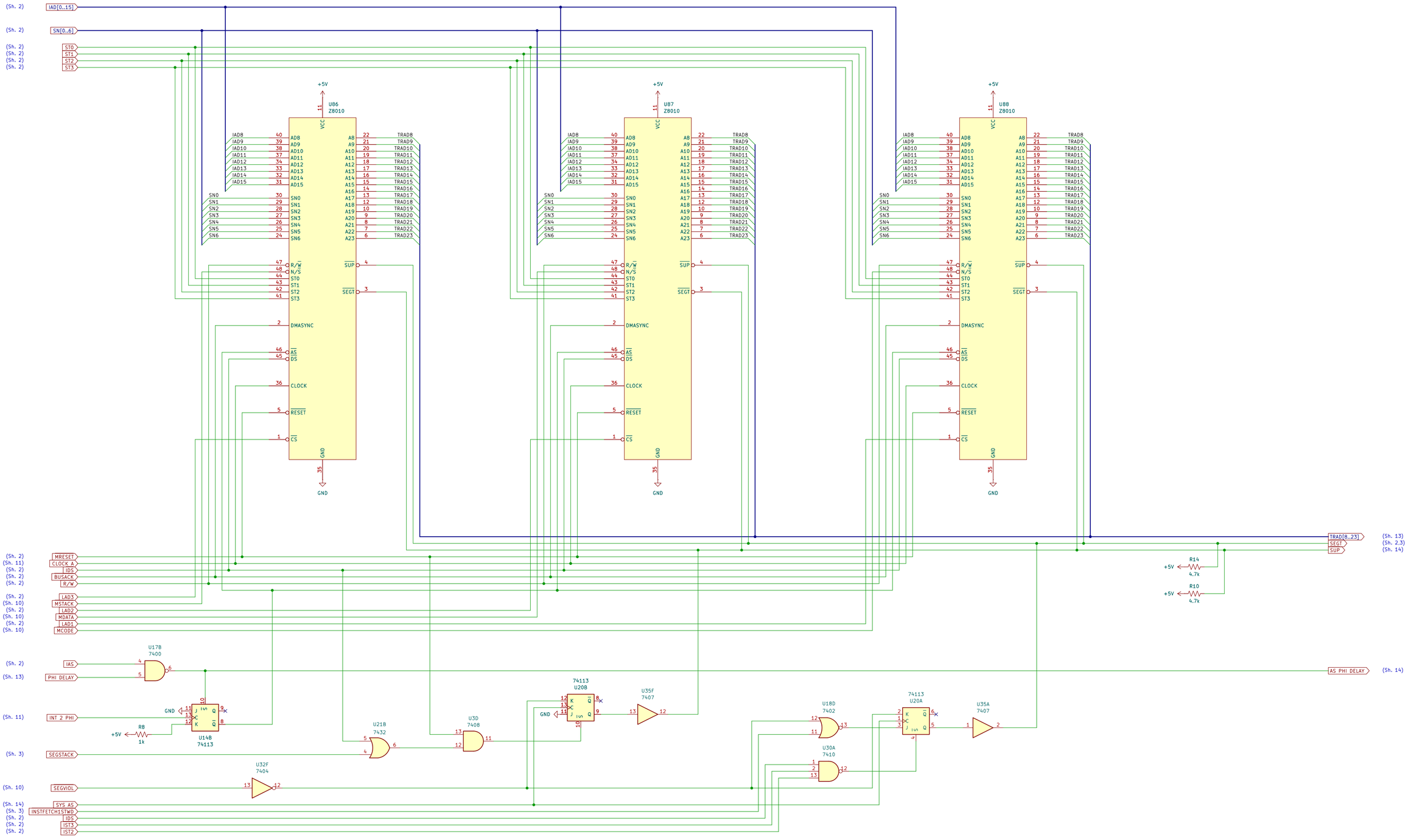
R6 1k +5V
P2 – 16C P2_PIO_D.D./ACKNOWLEDGE

PIO_0_IEO (Sh. 14)

Serial Input/Output: Baud Rate Generators
Parallel I/O Ports
Author: Oliver Lehmann <lehmann@ans-netz.de>
Sheet: /S8000 CPU 07/
File: S8000_CPU_07.kicad_sch

Title: Zilog S8000 CPU

Size: A2	Date: 2026-11-03	**Rev: 1.1**
KiCad E.D.A. 10.0.7		Id: 7/16



Memory Management Units

Author: Oliver Lehmann <lehmann@ans-netz.de>

Sheet: /S8000 CPU 08/

File: S8000_CPU_08.kicad_sch

Title: Zilog S8000 CPU

Size: A2	Date: 2026-11-03	**Rev: 1.1**
KiCad E.D.A. 10.0.7		Id: 8/16

(Sh. 2) ADIN[0..15]

(Sh. 2) LAD[0..15]

U74 2732 (1L) — +5V, VCC, GND, OE/VPP, CE; LAD1–LAD12 → A0–A11; D0–D7 → ADIN0–ADIN7

U75 2732 (2L) — +5V, VCC, GND, OE/VPP, CE; LAD1–LAD12 → A0–A11; D0–D7 → ADIN0–ADIN7

U59 2114 — +5V, VCC, GND, WE, CS; LAD1–LAD10 → A0–A9; DQ1–DQ4 → ADIN8–ADIN11

U57 2114 — +5V, VCC, GND, WE, CS; LAD1–LAD10 → A0–A9; DQ1–DQ4 → ADIN0–ADIN3

U76 2732 (1H) — +5V, VCC, GND, OE/VPP, CE; LAD1–LAD12 → A0–A11; D0–D7 → ADIN8–ADIN15

U77 2732 (2H) — +5V, VCC, GND, OE/VPP, CE; LAD1–LAD12 → A0–A11; D0–D7 → ADIN8–ADIN15

U60 2114 — +5V, VCC, GND, WE, CS; LAD1–LAD10 → A0–A9; DQ1–DQ4 → ADIN12–ADIN15

U58 2114 — +5V, VCC, GND, WE, CS; LAD1–LAD10 → A0–A9; DQ1–DQ4 → ADIN4–ADIN7

U56A 7432

(Sh. 10) BDMEMA2K → U56A pin 1

(Sh. 2) IW/R → U56A pin 2

U56A pin 3 → U56D pin 11

(Sh. 10) BDMEMB2K → U56D pin 12

U56D 7432 — pin 13 tied to pin 12

TP9

(Sh. 2) MREQ → U27B pin 5

(Sh. 2) IR/W → U27B pin 4

U27B 7432 — pin 6 → U59 pin 9, U57 pin 9, U60 pin 9, U58 pin 9

(Sh. 4) LAD0

U81F 7414 — pin 13 → pin 12

(Sh. 2) LAD0 → U54C pin 9, U54A pin 1

U54C 7427 — pin 11 tied to pin 10

U54A 7427 — pin 2 tied to pin 13

U56C 7432 — pin 9 → U54C pin 8; pin 10 → U27B pin 6; pin 8 → MEMC2KHI

(Sh. 2) IW/B → U54A pin 2

U54A pin 12 → U56B pin 4

(Sh. 10) BDMEMC2K → U56B pin 5

U56B 7432 — pin 6 → MEMC2KLO

Read-Only Memory
Read/Write Memory
Author: Oliver Lehmann <lehmann@ans-netz.de>

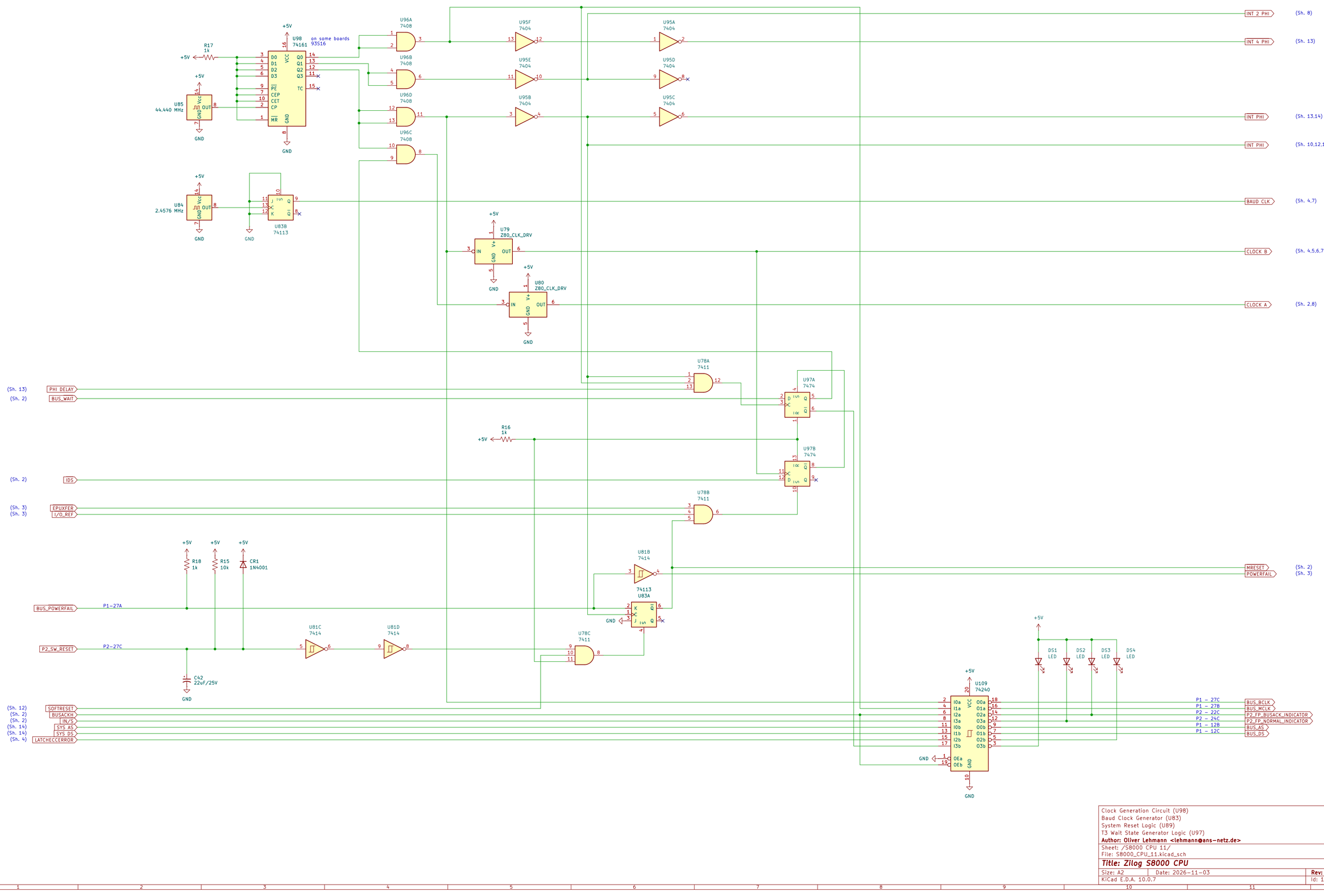
Sheet: /S8000 CPU 09/
File: S8000_CPU_09.kicad_sch

Title: Zilog S8000 CPU

Size: A3	Date: 2026-11-03	**Rev: 1.1**
KiCad E.D.A. 10.0.7		Id: 9/16

(Sh. 2) ISNAD6
(Sh. 2) ADIN[0..15]
(Sh. 2) LAD[0..15]
(Sh. 2) ISNAD6
(Sh. 2) ISNAD0
(Sh. 14) BDMEMON
(Sh. 2) MREQ
(Sh. 2) MREQ
(Sh. 14) BDMEMON
(Sh. 12) SELNBRKREG
(Sh. 11) INT PHI
(Sh. 2) DS
(Sh. 2) SN[0..6]
(Sh. 2) IW/R
(Sh. 12) SELSYSBAKREG
(Sh. 2) IR/W
(Sh. 14) SEG USER
(Sh. 2) IN/S
(Sh. 14) NONSEGUSER
(Sh. 2) IST2
(Sh. 2) IST3
LAD15
LAD13
LAD14
GND
+5V
U55 74138
U34B 74260
U23C 7410
U3C 7408
U21D 7432
U38 7408
U22D 7432
U38A 7408
U61B 74260
U5B 7474
U22A 7432
U18A 7402
U71 74373
U18C 7402
U72 74373
ADIN5
ADIN7
ADIN2
ADIN0
ADIN3
ADIN1
ADIN4
ADIN6
LE
OE
BRK5
BRK7
BRK2
BRK0
BRK3
BRK1
BRK4
BRK6
SN3
SN5
SN2
SN4
SN1
U73 74244
U53 7485
U52 7485
BRK[0..7]
LAD12
LAD13
LAD14
LAD15
LAD8
LAD9
LAD10
LAD11
DATA REF
1k
R11
U12A 74244
ADIN0
ADIN1
ADIN2
ADIN3
U1D 7404
U29E 7404
U16A 7451
U16B 7451
BDMEMA2K (Sh. 9)
BDMEMB2K (Sh. 9)
BDMEMC2K (Sh. 9)
IMEML (Sh. 12,13,14)
IMEMH (Sh. 14)
MANUAL_NMI (Sh. 3)
POWERFAIL (Sh. 3)
LATCHECCERROR (Sh. 4)
ENAONBDNMI (Sh. 3)
MSTACK (Sh. 8)
MDATA (Sh. 8)
NON SEGMENTED
SEGMENTED
E11
E12
E10
E3
E2
E1
E4
E5
E6
E7
E8
E9
U17C 7400
U15B 7402
U15C 7402
U29C 7404
U30B 7410
U2C 7410
U23A 7410
U29F 7404
U3A 7408
U17A 7400
R1
4.7k
MCODE (Sh. 8)
SEGVIOL (Sh. 8)

Memory Management Control Logic
Parity Error Checking: error buffer (U12)
Author: Oliver Lehmann <lehmann@ans-netz.de>
Sheet: /S8000 CPU 10/
File: S8000_CPU_10.kicad_sch
Title: Zilog S8000 CPU
Size: A2 | Date: 2026-11-03 | **Rev: 1.1**
KiCad E.D.A. 10.0.7 | Id: 10/16

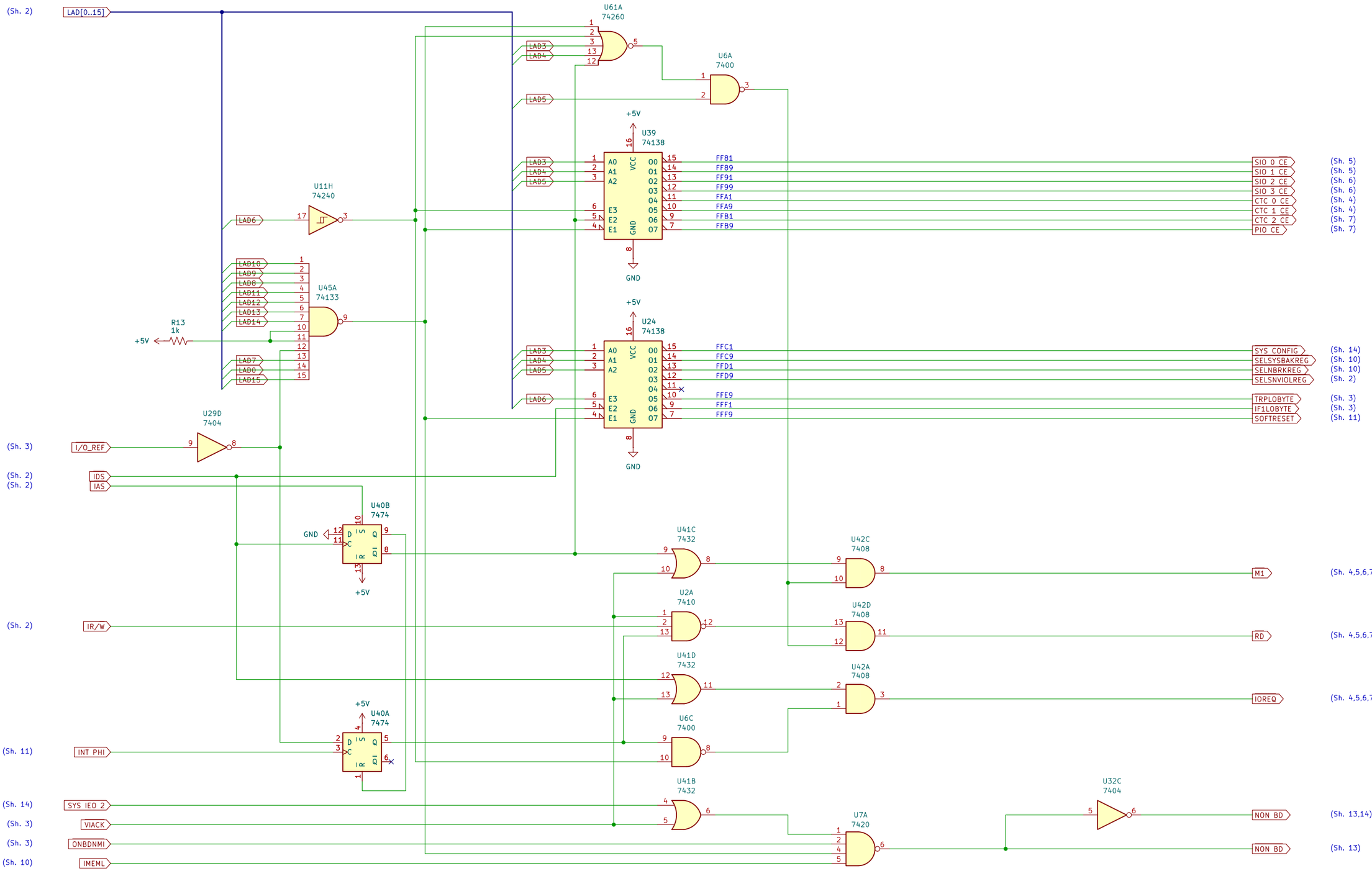


Clock Generation Circuit (U98)
Baud Clock Generator (U83)
System Reset Logic (U89)
T3 Wait State Generator Logic (U97)
Author: Oliver Lehmann <lehmann@ans-netz.de>

Sheet: /S8000 CPU 11/
File: S8000_CPU_11.kicad_sch

Title: Zilog S8000 CPU

Size: A2	Date: 2026-11-03	**Rev: 1.1**
KiCad E.D.A. 10.0.7		Id: 11/16



Peripheral Handshaking Logic (U24, U39)

Author: Oliver Lehmann <lehmann@ans-netz.de>

Sheet: /S8000 CPU 12/
File: S8000_CPU_12.kicad_sch

Title: Zilog S8000 CPU

Size: A3	Date: 2026-11-03	**Rev: 1.1**
KiCad E.D.A. 10.0.7		Id: 12/16

(Sh. 2) IAS

R7
1k
+5V

U13A
7474

(Sh. 11) INT PHI
(Sh. 11) INT 4 PHI

U13B
7474
GND

PHI DELAY (Sh. 8.11)

ADDR ENAL (Sh. 14)

(Sh. 12) NON BD

U30C
7410

BD_DATAEN (Sh. 2)

(Sh. 4) ENA BYTE SWAP

U31A
7420

(Sh. 2) IDS

U21C
7432

(Sh. 2) BUSACK

U33B
7411

(Sh. 3) MMU/EPUREF

U31B
7420

(Sh. 2) IW/R

U15D
7402

(Sh. 2) IDS

U29B
7404

(Sh. 14) MMUONH
(Sh. 10) IMEML
(Sh. 2) IST3

U23B
7410

U21A
7432

(Sh. 4) ENA BYTE SWAP
(Sh. 12) NON BD

U17D
7400

(Sh. 2) IAD[0..15]
(Sh. 2) LAD[0..15]
(Sh. 2) ISNAD[0..7]
(Sh. 13) TRAD[8..23]

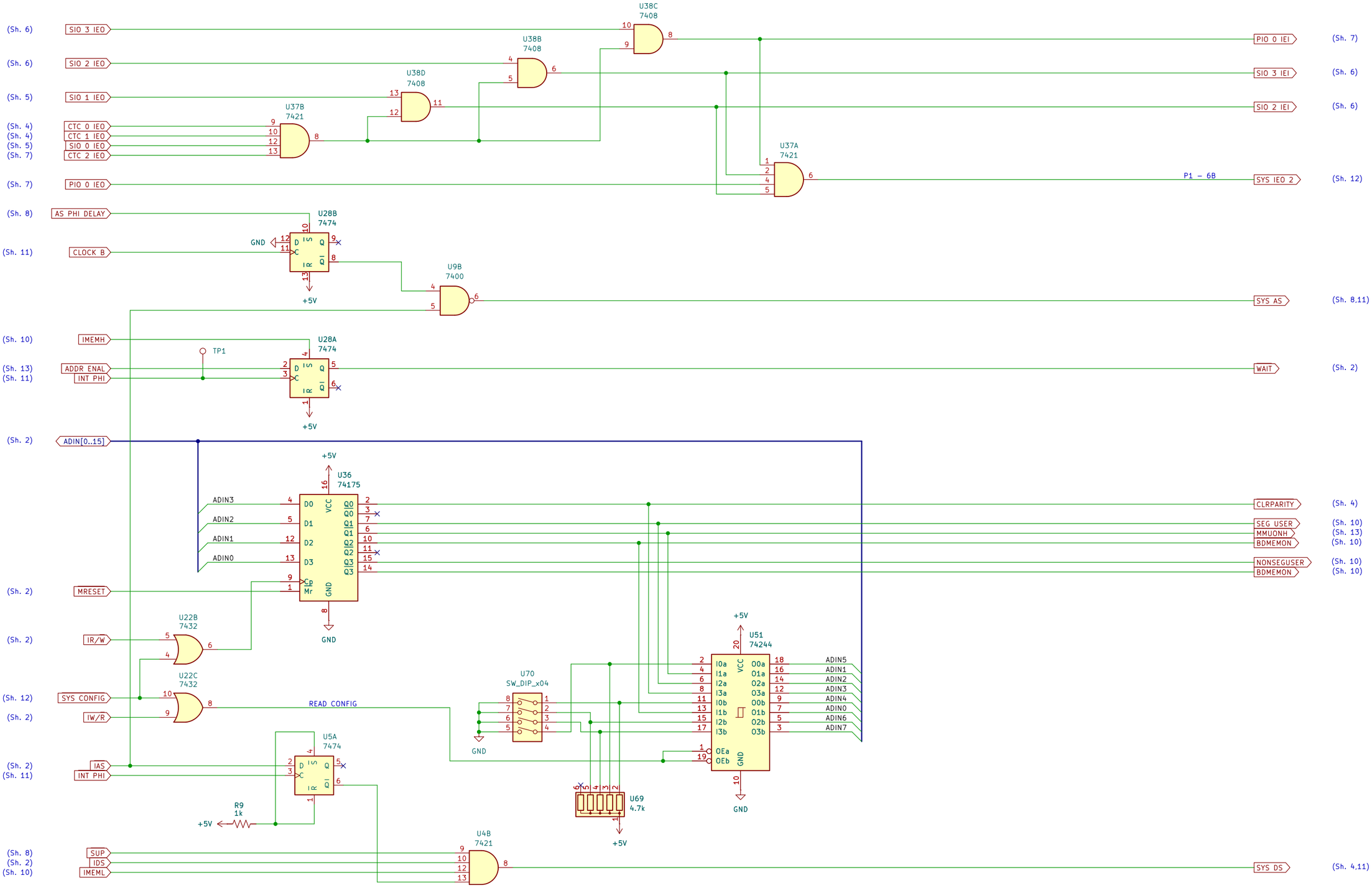
U114 74244
U117 74244
U116 74244
U115 74244
U110 74244
U113 74245
U111 74245
U112 74244

BUS_AD[0..23]

Address/Data Buffers and Steering Logic
Byte Swap Buffer (U112)
Author: Oliver Lehmann <lehmann@ans-netz.de>
Sheet: /S8000 CPU 13/
File: S8000_CPU_13.kicad_sch

Title: Zilog S8000 CPU

Size: A2	Date: 2026-11-03	**Rev: 1.1**
KiCad E.D.A. 10.0.7		Id: 13/16

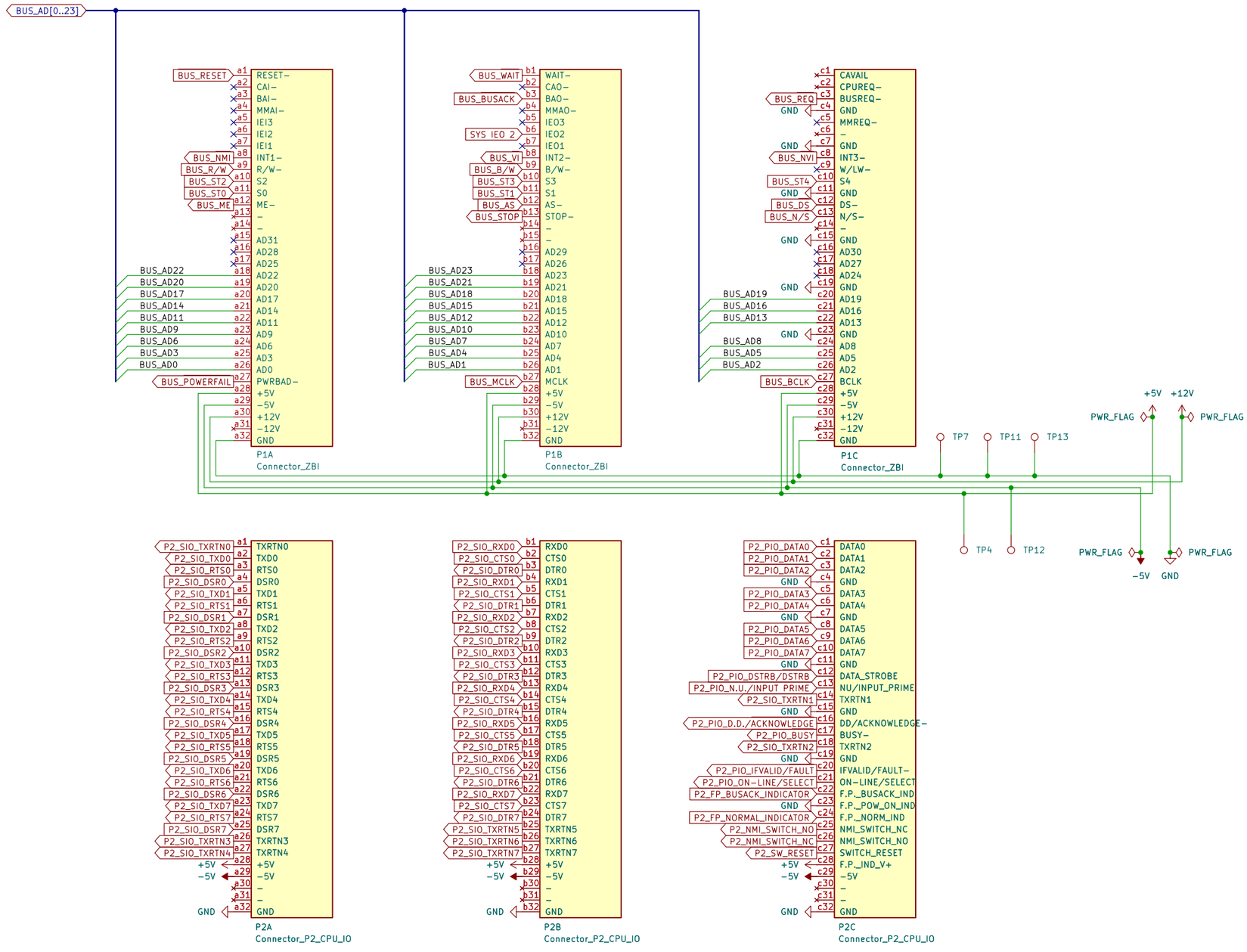


Interrupt Priority Connection
System Configuration Register
T2 Wait State Generator Logic (U28)
Author: Oliver Lehmann <lehmann@ans-netz.de>

Sheet: /S8000 CPU 14/
File: S8000_CPU_14.kicad_sch

Title: Zilog S8000 CPU

Size: A3	Date: 2026-11-03	**Rev: 1.1**
KiCad E.D.A. 10.0.7		Id: 14/16



Connectors

Author: Oliver Lehmann <lehmann@ans-netz.de>

Sheet: /S8000 CPU 15/

File: S8000_CPU_15.kicad_sch

Title: Zilog S8000 CPU

Size: A3	Date: 2026-11-03	**Rev: 1.1**
KiCad E.D.A. 10.0.7		Id: 15/16